Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	Slot: L15+L16
Regno: *23BCE1344*	Name: *Aditya*

EX4

RSA - CRYPTOGRAPHIC ALGORITHM

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Code: (--CyptoFunctions--)

Cypto_functions.py:

```
# crypto_functions.py
import random
import math

def is_prime(n, k=40):
    if n <= 1 or n % 2 == 0:
        return False
    if n <= 3:
        return True
    r, d = 0, n - 1
    while d % 2 == 0:
        d //= 2
        r += 1
    for _ in range(k):
        a = random.randrange(2, n - 2)
        x = pow(a, d, n)
        if x == 1 or x == n - 1:
            continue
        for _ in range(r - 1):
            x = pow(x, 2, n)
            if x == n - 1:
                break
        else:
            return False
    return True

def generate_prime(bits=1024):
    while True:
        p = random.getrandbits(bits)
        p |= (1 << bits - 1) | 1
        if is_prime(p):
            return p

# Extended Euclidean Algorithm
def egcd(a, b):
    if a == 0:
        return b, 0, 1
    g, y, x = egcd(b % a, a)
    return g, x - (b // a) * y, y
def modinv(e, phi):
    g, x, _ = egcd(e, phi)
    if g != 1:
        raise Exception("Modular inverse does not exist")
    return x % phi

def generate_rsa_keys():
    p = generate_prime(1024)
    q = generate_prime(1024)
    n = p * q
    phi = (p - 1) * (q - 1)
    e = 65537
    d = modinv(e, phi)
    return p, q, n, phi, e, d
```

```python
def rsa_encrypt(message, e, n):
    m = int.from_bytes(message.encode(), 'big')
    c = pow(m, e, n)
    return c
def rsa_decrypt(cipher, d, n):
    m = pow(cipher, d, n)
    msg_len = (m.bit_length() + 7) // 8
    return m.to_bytes(msg_len, 'big').decode()

# DEMO RUN
def main():
    message = "Hello RSA"
    print("Message:", message)
    p, q, n, phi, e, d = generate_rsa_keys()
    print("\np value:", p)
    print("\nq value:", q)
    print("\nn = p * q:", n)
    print("\nphi(n):", phi)
    print("\ne:", e)
    print("\nd:", d)
    cipher = rsa_encrypt(message, e, n)
    print("\nEncrypted Text:", cipher)
    decrypted = rsa_decrypt(cipher, d, n)
    print("\nDecrypted Text:", decrypted)
if __name__ == "__main__":
    main()
```

OUTPUT:

```
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python crypto_functions.py
Message: Hello RSA

p value: 16640911145193191712944044205942667750461884470242144356940120275588869609032577993032334405427638060599971700085349520635029857068210202776144890855197768609645205542777025792827026606958438126231454812786912340239665590478568322612693876158973590972657872102095920810793255791048507256106635275721953120290 3

q value: 12685040595850158153697306999696385957184647285019623754666717379803543284915922182820335391950086813448607759252016266965445757763798734602837192107360033426119478034599427825843412554977256562039117699694267408886328584424821559278813877400410061961973292096658951361255440621167138603522449718825219348085 7

n = p * q: 211090633428710982291351170233577510096535611288131959180713919154114407637671708688885722822147747417150711034158713441697867233334759767936579243572242737096086017329464322408262696676287792987509697413480124837281461227466580400959097872856000575277401466439933867598405680313598949738685768198393833326186438368389424467848819206335765755680857036917334749031331376097293933468236630255856845361004473594994157239224101769575742774502686518662534719772655667423794878753597787190125768021976074134754744223817507193238046430269890873335343323832531704432402535161459927937949786513342003109196339477298319133278871

phi(n): 211090633428710982291351170233577510096535611288131959180713919154114407637671708688885722822147747417150711034158713441697867233334759767936579243572242737096086017329464322408262696676287792987509697413480124837281461227466580400959097872856000575277401466439933867598405680313598949738685768198393833326183505773215320132862155071215201850310092383741805722201706923317546921788417802382715783812667011075892992932903651908156952130194856249246365114763998872202184103958601418282587241057825046659276869725693892183254470127795518851461925484765933391389394187415844407207449168721204385232334309778436018864411 2

e: 65537

d: 107460164534264377407973344713716625255210302568105750250181706284064253505891957474209902353103060821801412518617531418212093084910014039972352919775100087549578372463843602674929709159878994116945939466346299112657298792010124386486997914645082124494254316261585266714765837806164483576174882652593397031622752080683930079080215445335501767427493052760699442986123179398878981264975484292681304752036268810786300751057363838155302186428568578121681613933917167971194086180891434026827529675469150299362885400092047736255731704035341708411010909788738934856530445816492694138673003976449245318684666031612617143557 61

Encrypted Text: 152244256683020422369060417003961153864331634310596032866842055101994326304364598432312277237892984289240228050639794158984927314193312607812990206667108354540996870739763187625307143881018970957490405490129169608474685586567278194400403518591139747722045901691035693744208809847086097893247038092453227816234682010121021444932455741874717859538906258030945158597259642224579567246412773643608647527854717072649646894675162405590346046004079143487381520835141104346179010665028392508229615418888126143711520170601723862147243921251795857774189915618908623001481205422942753412663244481493408023855989552517592705214 56

Decrypted Text: Hello RSA
(lab) ak@1125:~/Desktop/Data/CryptoLab$
```

Sender (server.py):

```
import socket
import crypto_functions as cf

HOST = "127.0.0.1"
PORT = 12345

print("Generating RSA keys (this may take some time)...")
p, q, n, phi, e, d = cf.generate_rsa_keys()
print("\np value:", p)
print("\nq value:", q)
print("\nn = p * q:", n)
print("\nphi(n):", phi)
print("\ne:", e)
print("\nd:", d)

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)

print("\nServer ready (RSA Encryption)")
conn, addr = server.accept()
print("Connected:", addr)

while True:
   msg = input("\nEnter Message (or exit): ")
   if msg.lower() == "exit":
       conn.send(b"exit")
       break

   cipher = cf.rsa_encrypt(msg, e, n)
   print("\nEncrypted Text:", cipher)
   payload = f"{cipher}".encode()
   conn.send(payload)

conn.close()
server.close()
```

Receiver (client.py):

```
import socket
import crypto_functions as cf

HOST = "127.0.0.1"
PORT = 12345

print("Enter RSA private key details from sender output")
n = int(input("Enter n: "))
d = int(input("Enter d: "))

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
print("\nConnected to server (RSA Decryption)")

while True:
   data = client.recv(4096).decode()
   if data == "exit":
       break

   cipher = int(data)
   print("\nReceived Encrypted Text:", cipher)

   plaintext = cf.rsa_decrypt(cipher, d, n)
   print("Decrypted Text:", plaintext)

client.close()
```

Output:

```
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python server.py
Generating RSA keys (this may take some time)...

p value: 15492803691397635519653232228295992134834913104537520027753495637930847473976871295
80208398442622555995114454433779212230755484356163886307639444401573486236558075165078204304698
36343459269171928497307921112566176846579035170052442221241459427019522677823113272877593252781
699424230748490445062706612194437979

q value: 15535626155229908726881946221942896923455076924822043993267366940845194079768385990867
39278341582413363508136754900593522172801831983214354212460574833703285648929574202247713282 2
83846173007199121962940684364093902757805707719481614177266480980862633020649939526455696561 20
526198412222778662329071034889991511

n = p * q: 24069040624591958565680291836087108086621106429252170608855740741320455429228548715
47503275682388140549402745454282375168552100883910124218882418975640474686548378113506297976 6
33117955365496839118970242420227648853889448088431856317739903234864541777487092775388156906 25
938429373437670372247297910161880077660639057556982503048120304122550570854603044007309802520 5
984874608178762105701398540540680002404072386309810707386938590799954682637158501014482201472 9
978645565639777020525480200260591680610983601818156251481910426696191660290414090426273582281 4
001332678158397299473393292859528585668748556033501916875996269

phi(n): 24069040624591958565680291836087108086621106429252170608855740741320455429228548715
47503275682388140549402745454282375168552100883910124218882418975640474686548378113506297976 6
33117955365496839118970242420227648853889448088431856317739903234864541777487092775388156906 25
938429373437670372247297910161880077660639057556982503048120304122550570854603044007309802520 5
984874608178762105701398540540680002404072386309810707386938590799954682637158501014482201472 9
978645565639777020525480200260591680610983601818156251481910426696191660290414090426273582281 4
001332678158397299473393292859528585668748556033501916875996269

d: 14288938516883582280466349000818485959475828735567911284903952042701908225658714083235227 42
06897897957453844792619400395300800553746782916975763046078588270538709087578343460186871800 70
63804241145615110886741530942762526140617694178485903253366609284734867813009486276075856180 04
53536085138627892608857327643977487631769348670536227243317916688839762575510034015793073630 35
92782185625044920913295170430708368230870009222478421765609367703712645587449030439426952547 99
91728879346591410752510637710985202334143961304696090005824538358952189779575156731469174181 58
53403418128330772415489944407831558257696609516494628053

Server ready (RSA Encryption)
Connected: ('127.0.0.1', 48426)

Enter Message (or exit): Hello

Encrypted Text: 13717297407117844825405453745178810969098574991979788565896942703124097314818 0
95066071453217732017585416810433146014717947506734795604447940510019547638795018726051724319 90
58281503168937694953717025079936656570456255857218036077448787858647711831017836911015352428 7
95489670261417198773300219545654158397179954267496976871809901788441095237457917501863444525 15
73626877064170222536766221669664377372289911198108232920813671032595726468975803056327039745 05
68400207213270347348830683303353696509627433103974666688962136284285123624444928211515384143 92
64156640603760835494824145240332636797604937366138486234525286976731 2

Enter Message (or exit): How are you?

Encrypted Text: 92882801380029512262085363776258279400812665882860587700581345783266323231355 9
31730326402074010350160388372292020204707626848093695227780705108976030154933728236832042185 52
85608808004151319833420574597340420057312366299108043952172394300619499670672818764102382816 11
36393954376504259959769446508737481443776120220695252940519464662751761704154117663226810006 17
78339583602913866113586652981453110113136488746438995730390253764142670258689617830631310076 18
48554748004131971995372242061006319371092833839512343363191621852099250263783453568342489817 25
85445336762833649195167002128617045698621093037051313792902465899779

Enter Message (or exit): What do you do?
```

```
(lab) ak@1125:~/Desktop/Data/CryptoLab$ python client.py
Enter RSA private key details from sender output
Enter n: 24069040624591958565680291836087108086621106429252170608855740741320455429228548715 47
50327568238814054994027454542823751685521008839101242188824189756404746865483781135062979676 33
11795536549683911897024242022764885388944808843185631773990323486454177748709277538815690625 9
8342937343767037224729791016188007766063905755698250304812030412255057085460304400730980252 05
9848746081787621057013985405406800024040723863098107073869385907999546826371585010144822014 729
9786455656397770205254802002605916806109836018181562514819104266961916602904140904262735822 814
0013326781583972994733932928595285856687485560335019168725996269

phi(n): 24069040624591958565680291836087108086621106429252170608855740741320455429228548715
47503275682388140549402745454282375168552100883910124218882418975640474686548378113506297976 6
33117955365496839118970242420227648853889448088431856317739903234864541777487092775388156906 25
938429373437670372247297910161880077660639057556982503048120304122550570854603044007309802520 5
984874608178762105701398540540680002404072386309810707386938590799954682637158501014482201472 9
978645565639777020525480200260591680610983601818156251481910426696191660290414090426273582281 4
001332678158397299473393292859528585668748556033501916875996269

e: 65537

d: 14288938516883582280466349000818485959475828735567911284903952042701908225658714083235227 42
06897897957453844792619400395300800553746782916975763046078588270538709087578343460186871800 70
63804241145615110886741530942762526140617694178485903253366609284734867813009486276075856180 04
53536085138627892608857327643977487631769348670536227243317916688839762575510034015793073630 35
92782185625044920913295170430708368230870009222478421765609367703712645587449030439426952547 99
91728879346591410752510637710985202334143961304696090005824538358952189779575156731469174181 58
53403418128330772415489944407831558257696609516494628053

Server ready (RSA Encryption)
Connected: ('127.0.0.1', 48426)

Enter Message (or exit): Hello

Encrypted Text: 13717297407117844825405453745178810969098574991979788565896942703124097314818 0
95066071453217732017585416810433146014717947506734795604447940510019547638795018726051724319 90
58281503168937694953717025079936656570456255857218036077448787858647711831017836911015352428 7
95489670261417198773300219545654158397179954267496976871809901788441095237457917501863444525 15
73626877064170222536766221669664377372289911198108232920813671032595726468975803056327039745 05
68400207213270347348830683303353696509627433103974666688962136284285123624444928211515384143 92
64156640603760835494824145240332636797604937366138486234525286976731 2

Enter Message (or exit): How are you?

Encrypted Text: 92882801380029512262085363776258279400812665882860587700581345783266323231355 9
31730326402074010350160388372292020204707626848093695227780705108976030154933728236832042185 52
85608808004151319833420574597340420057312366299108043952172394300619499670672818764102382816 11
36393954376504259959769446508737481443776120220695252940519464662751761704154117663226810006 17
78339583602913866113586652981453110113136488746438995730390253764142670258689617830631310076 18
48554748004131971995372242061006319371092833839512343363191621852099250263783453568342489817 25
85445336762833649195167002128617045698621093037051313792902465899779

Enter Message (or exit): What do you do?

(lab) ak@1125:~/Desktop/Data/CryptoLab$ python client.py
Enter RSA private key details from sender output
Enter n: 24069040624591958565680291836087108086621106429252170608855740741320455429228548715 47
50327568238814054994027454542823751685521008839101242188824189756404746865483781135062979676 33
11795536549683911897024242022764885388944808843185631773990323486454177748709277538815690625 9
8429373437670372247297910161880077660639057556982503048120304122550570854603044007309802520 598
4874608178762105701398540540680002404072386309810707386938590799954682637158501014482201472 997
8645565639777020525480200260591680610983601818156251481910426696191660290414090426273582281 400
1332678158397299473393292859528585668748556033501916875996269
Enter d: 14288938516883582280466349000818485959475828735567911284903952042701908225658714083 23
52274206897897957453844792619400395300800553746782916975763046078588270538709087578343460186 87
1800706380424114561511088674153094276252614061769417848590325336660928473486781300948627607 585
6180045353608513862789260885732764397748763176934867053622724331791668883976257551003401579 307
3630359278218562504492091329517043070836823087000922247842176560936770371264558744903043942 695
2547999172887934659141075251063771098520233414396130469609000582453835895218977957515673146 917
4181585340341812833077241548994440783155825769660951649462805 3

Connected to server (RSA Decryption)

Received Encrypted Text: 13717297407117844825405453745178810969098574991979788565896942703124 0
97314818095066071453217732017585416810433146014717947506734795604447940510019547638795018726 05
17243199058281503168937694953717025079936656570456255857218036077448787858647711831017836911 01
53552428795489670261417198773300219545654158397179954267496976871809901788441095237457917501 86
34445251573626877064170222536766221669664377372289911198108232920813671032595726468975803056 32
70397450568400207213270347348830683303353696509627433103974666688962136284285123624444928211 51
53841439264156640603760835494824145240332636797604937366138486234525286976731 2
Decrypted Text: Hello

Received Encrypted Text: 92882801380029512262085363776258279400812665882860587700581345783266 3
23231355931730326402074010350160388372292020204707626848093695227780705108976030154933728236 83
20421855285608808004151319833420574597340420057312366299108043952172394300619499670672818764 10
23828161136393954376504259959769446508737481443776120220695252940519464662751761704154117663 22
68100061778339583602913866113586652981453110113136488746438995730390253764142670258689617830 63
13100761848554748004131971995372242061006319371092833839512343363191621852099250263783453568 34
24898172585445336762833649195167002128617045698621093037051313792902465899779
Decrypted Text: How are you?

Received Encrypted Text: 10450100981266554270555644157409427993412760423227458568659302175131 5
70418009923808316338700573747152842410506569867564074499729197093269367627828870683415265968 60
89425826421301453443782715974611011073633777394362853055448441706520781444026146569038933176 70
61573824776919106315909909492087074378358136696704426560956162510237398313018745468567621850 00
68148410492230837742637767967011799345245078354462254979034309414335015979756621774257810513 92
73544769546633782743738680086038377465600018707442323917547512986979384863431593988113349163 94
89326620870067239676516323296419369173767850217598952854259687488580972271 0817
Decrypted Text: What do you do?
```